1 Part A — User and System Investigation

1.1 Determine the Username of the Account Currently Being Used

Use the following command:

```
whoami
```

The `whoami` command displays the username of the currently logged-in user.

1.2 Display UID, Primary GID and Supplementary Groups

Use the following command:

```
id
```

This command displays the User ID (UID), primary Group ID (GID), and supplementary group memberships.

1.3 Display All Users Currently Logged In

Use the following command:

```
who
```

This command displays information about all users currently logged in to the system.

1.4 Determine the Activities of Currently Logged-In Users

Use the following command:

```
w
```

This command displays logged-in users and information about their current activity.

1.5 Examine the Current Account in `/etc/passwd`

Use the following command:

```
grep "^$(whoami):" /etc/passwd
```

From the output, the following information is identified:

Information	Value Obtained
Username	ananya
UID	1000
GID	1000
Home Directory	`/home/ananya`
Login Shell	`/bin/bash`

Table 1: Current User Account Information

1.6 Determine the Number of User-Account Entries

Use:

```
wc -l /etc/passwd
```

Each line in `/etc/passwd` represents one user account entry.

1.7 Display Only the Username and UID Fields

Use:

```
cut -d: -f1,3 /etc/passwd
```

The first field contains the username and the third field contains the UID.

1.8 Find All Accounts Whose UID is 0

Use:

```
awk -F: '$3 == 0 {print $1}' /etc/passwd
```

UID 0 represents superuser privileges. Any account with UID 0 has unrestricted administrative access and can perform any operation on the system. Therefore, accounts with UID 0 must be strictly controlled to prevent unauthorized access.

1.9 Compare `/etc/passwd` and `/etc/shadow`

The following table compares the two files:

`/etc/passwd`	`/etc/shadow`
Contains general information about user accounts.	Contains password hashes and password aging information.
Normally readable by all users and system programs.	Accessible only to privileged users, such as root.
Contains username, UID, primary GID, home directory, and login shell.	Contains encrypted password hashes and password expiry information.
The password field usually contains an `x`, indicating that the password hash is stored elsewhere.	Stores the actual protected password hash.

Table 2: Comparison Between /etc/passwd and /etc/shadow

Password hashes are stored in `/etc/shadow` because `/etc/passwd` is normally readable by regular users. Restricting access to password hashes improves system security and reduces the risk of unauthorized password-cracking attempts.

2 Part B — Creating and Verifying Users

2.1 Create the Required Users

Create the following users:

- alice
- bob
- charlie

Use:

```
sudo useradd -m alice
sudo useradd -m bob
sudo useradd -m charlie
```

The `-m` option creates a home directory automatically for each user.

2.2 Verify That the Accounts Have Been Added

Use:

```
grep -E '^(alice|bob|charlie):' /etc/passwd
```

The command verifies that all three user accounts have been successfully added to the system.

2.3 Determine the UID Assigned to Each User

Use:

```
id -u alice
id -u bob
id -u charlie
```

The obtained UIDs are:

Username	UID
alice	1001
bob	1002
charlie	1003

Table 3: UID Assigned to Newly Created Users

2.4 Determine Alice's Home Directory and Default Shell

Use:

```
grep "^alice:" /etc/passwd
```

The following information is obtained:

Information	Value Obtained
Home Directory	`/home/alice`
Default Shell	`/bin/sh`

Table 4: Alice Account Information

2.5 Inspect the `/home` Directory

Use:

```
ls -l /home
```

The following home directories should be present:

- `/home/alice`
- `/home/bob`
- `/home/charlie`

2.6 Set Passwords for All Three Accounts

Use:

```
sudo passwd alice
sudo passwd bob
sudo passwd charlie
```

Enter and confirm a password for each account.

2.7 Switch to Alice Using a Login Shell

Use:

```
su - alice
```

Verify the account change:

```
whoami
```

The output is:

```
alice
```

2.8 Return to the Original Account

Use:

```
exit
```

Verify the current user:

```
whoami
```

The output should display the original username.

3 Part C — Group Management and Access Control

3.1 Create the Required Groups

Create the following groups:

- security
- developers

Use:

```
sudo groupadd security
sudo groupadd developers
```

3.2 Add Alice and Charlie to the Security Group

Use:

```
sudo usermod -aG security alice
sudo usermod -aG security charlie
```

This adds both `alice` and `charlie` to the `security` group.

3.3 Add Bob to the Developers Group

Use:

```
sudo usermod -aG developers bob
```

This adds `bob` to the `developers` group.

3.4 Verify the Group Membership of All Users

Without switching accounts, use:

```
id alice
id bob
id charlie
```

Alternatively:

```
groups alice
groups bob
groups charlie
```

The expected group membership is:

User	Supplementary Group
alice	security
bob	developers
charlie	security

Table 5: Initial Group Membership

3.5 Display the Security and Developers Group Entries

Use:

```
grep -E '^(security|developers):' /etc/group
```

This displays the group entries for `security` and `developers`.

3.6 Add Bob to the Security Group

Use:

```
sudo usermod -aG security bob
```

The `-aG` option appends the `security` group to Bob's existing supplementary groups. Therefore, Bob remains a member of the `developers` group.

3.7 Verify That Bob Belongs to Both Groups

Use:

```
groups bob
```

Alternatively:

```
id bob
```

The final group membership of Bob is:

User	Groups
bob	developers, security

Table 6: Final Group Membership of Bob